

Brian Li

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Education

California Polytechnic State University, San Luis Obispo, Expected May 2028
Bachelor of Science in Computer Engineering – San Luis Obispo, CA
Adobe Ambassador, Cal Poly Engineering Ambassador, Hack4Impact Developer, BMES Member

Experience

Boeing, Software Test Engineer (Part-Time) – San Luis Obispo, CA Aug 2026 – present

- Automated 6 recurring regression test suites using Python and pytest, reducing manual testing time by ~60%.
- Validated 11 functional, boundary, error, and regression scenarios against software requirements.

Booz Allen Hamilton, Systems Engineer Intern – Honolulu, HI June 2026 – Aug 2026

- Architected a Python multi-vehicle simulation engine coupling flight control software with high-fidelity 3D physics and live telemetry for risk-free testing of drone fleet behaviors before hardware deployment.
- Implemented setpoint streaming and multi-vehicle coordination logic through Pegasus and Isaac Sim APIs to execute deterministic fleet experiments.
- Engineered typed telemetry, integrity checks, and automated diagnostics for reproducible simulation analysis.
- Built digital-twin environments using Blender and Cesium ion pipelines for realistic sensor and terrain simulation.
- Traced 47 system requirements and 48 risks to 9 simulation test scenarios for 100% verification coverage.

Operation SURF, Software Engineer (Pro Bono) – San Luis Obispo, CA Sept 2025 – May 2026

- Engineered scheduling workflows for 3,600 veterans and volunteers using Next.js, REST APIs, and MongoDB to streamline 2,500 annual program shift bookings with 100% data integrity.
- Consolidated shift and event queries into unified API endpoints to cut duplicate client-side fetches by ~65%.
- Saved 40% (\$30/month) in projected database costs by optimizing MongoDB aggregation pipelines and indexes.

Cal Poly Information Technology Services, IT Assistant (Part-Time) – CA Nov 2025 – May 2026

- Diagnosed and resolved 40+ tickets involving campus-network failures, DNS resolution, and application-access issues.

Projects

Balance-Training System (Hardware & Software Lead | 2nd Place, Cal Poly Hackathon)

- Led a \$74 balance-training prototype, interfacing 4 force sensors with Arduino to stream 20 Hz data to a React app.
- Calibrated and normalized sensor readings into left-right weight distribution, balance scores, and progress tracking.

AI Teaching VS Code Extension (Software Developer)

- Built an adaptive tutoring VS Code extension with code analysis, explanations, and gamification using Gemini API.
- Validated Gemini 2.5 Flash responses against Zod contracts and protected credentials with SecretStorage.

FPGA/RISC-V Embedded Game (Embedded Hardware & Software Developer)

- Co-engineered a real-time FPGA game on an OTTER MCU and Digilent Basys 3 using Assembly and SystemVerilog.
- Interfaced 10 switches, LEDs, and a 4-digit 7-segment display via memory-mapped I/O and custom peripheral logic.

Skills

Programming: Python, TypeScript, JavaScript, Java, C, C++, SQL, RISC-V Assembly, Bash

Frameworks & Data: React, Next.js, MongoDB, Mongoose, Supabase, SQLite, REST APIs, Zod, JSON Schema

Testing & Systems: pytest, GitHub Actions, Postman, Ruff, mypy, Docker, Git, Linux

Tools: Figma, Claude Code, Codex, Cursor, Gemini API, Ollama, DevOps, Microsoft Office, Excel, Word